SUSTAINABLE, RELIABLE AND EFFICIENT FLOATING PHOTOVOLTAIC POWER PLANTS

D1.1 Monitoring Concepts

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WP 1

SUSTAINABILITY

Deliverable D1.1: Monitoring Concepts

Identifying FPV interactions with ecosystems and establishing environmental impact guidelines

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Executive SuRE project summary

Floating PV, if it is to aid the transition to a climate-neutral and resilient society and contribute towards the EU policy goals, must overcome 3 challenges that are also highlighted in the Work Programme. FPV must prove its sustainability, by demonstrating low impact on biodiversity and satisfy end-of-life requirements, its longevity and reliability by demonstrating system components that satisfy structural and functional requirements for the entire lifecycle, and its affordability, by reducing the LCOE from FPV power plants. These are the challenges that the objectives of SuRE seek to overcome. Activities are structured into 3 generalizable topics, SUstainability, Reliability, and Efficiency, which gives SuRE FPV its name, and are designed to advance the entire FPV industry. We will further work with concrete technology developments for 3 leading European FPV technologies to improve their design, sustainability, cost competitiveness and application range. The three FPV technology providers are Ciel et Terre (CTI), who have installed 650 MW globally, Zimmermann PV-Steel Group (ZIM), who is dominating the European FPV market, and Sunlit Sea (Sunlit) who is providing a innovative FPV solution for off-shore deployment. CTI has recently prototyped a new floater design, which will be developed and tested in SuRE, first 50 kW, then on 5 MW scale. ZIM aims to expand their technology to higher sea states, and will build a 5 MW based on the developments in floater-, connection- and anchoring- technology in SuRE. Sunlit are about to scale up their FPV technology and see potential for large reduc- tions in cost and CO2 footprint through the activities planned in SuRE. They will build a smaller, but still commercially relevant, pilot of 100 on the Norwegian cost. Ultimately, SuRE will provide both cost-efficient and sustainable new FPV technologies and generalizable knowledge, thereby expanding the potential application areas without environmental sacrifices.

Executive deliverable summary

Research on the environmental impact of floating photovoltaic (FPV) systems is still in its infancy. This knowledge gap is hindering the permitting process by authorities responsible for environmental protection. More data and their analysis are needed to understand the environmental impact in relation to nature's needs, legal obligations and demands set by local authorities or other stakeholders using the water as a resource.

This document guides the reader in choosing the right monitoring variables, as well as setting up a cost-efficient and effective monitoring plan. Furthermore, it helps creating consistency between datasets, which may then together form one large and internally consistent database. Creating such a database is important for a relatively new technology like FPV. With more knowledge, certain aspects of the relations between FPV design and impact on water types will transpire. This way the sector will learn how to balance FPV impact and performance.

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1 Introduction

Research on the environmental impact of floating photovoltaic (FPV) systems is still in its infancy. With only a few fully commercial FPV plants installed to date, the limited amount of data and accompanying analysis is understandable. This knowledge gap is hindering, however, the permitting process by authorities responsible for environmental protection. More data and their analysis are needed to understand the environmental impact in relation to nature's needs, legal obligations and demands set by local authorities or other stakeholders using the water as a resource.

One essential data source is in situ monitoring data, which consists of the sampling of local water constituents and conditions. Other data sources are remote measurements or modelling, each having specific advantages, but they heavily rely on in situ data for their calibration and validation. This advice gives recommendations on how, when and what in situ data to collect. Goal of the proposed 'Monitoring advice' is to collect data which helps to understand the impact of FPV on a local water resource and its depending flora and fauna at a local scale.

This advice is part of the EU-HORIZON project SuRE-PV and is intended to be used within this project as well as within other FPV initiatives. The advice is written by Deltares and Fraunhofer ISE and is strongly based on the currently known relations within the natural system and the expected impacts of the FPV system on that system. The advice is set up as generic as possible, making it applicable to a range of different waterbodies as well as to different FPV systems and designs. The below monitoring recommendations will address the following questions, focusing mainly on the first two:

1. How does the FPV impact hydrodynamics? This includes impacts on surface waves, currents and stratification. Stratification is the separation of a body of water into distinct and stable vertical layers based on the density of the water (Figure 1).
2. How does FPV affect the lower tropic levels of the food web? The lower trophic levels include primary producers such as algae and aquatic vegetation, which utilize sunlight to produce oxygen and organic material. This directly provides food for the primary consumers such as zooplankton and macrofauna (e.g., larvae, snails or worms) and is thus essential for the productivity and energy flow in the ecosystem (Figure 2).
3. How does FPV affect the higher trophic levels of the food web? The artificial island formed by FPV may provide habitat for sessile organisms, a resting place for birds, shelter for fish, and attract or repel other organisms such as bats and insects (Figure 2).

The above three questions are not separated, on the contrary: the hydrodynamics in a waterbody may strongly affect the lower trophic food web, which in turn may strongly affect the higher trophic food web (and vice versa). Hence, the same indicators and monitoring requirements could be relevant for answering several of these questions.

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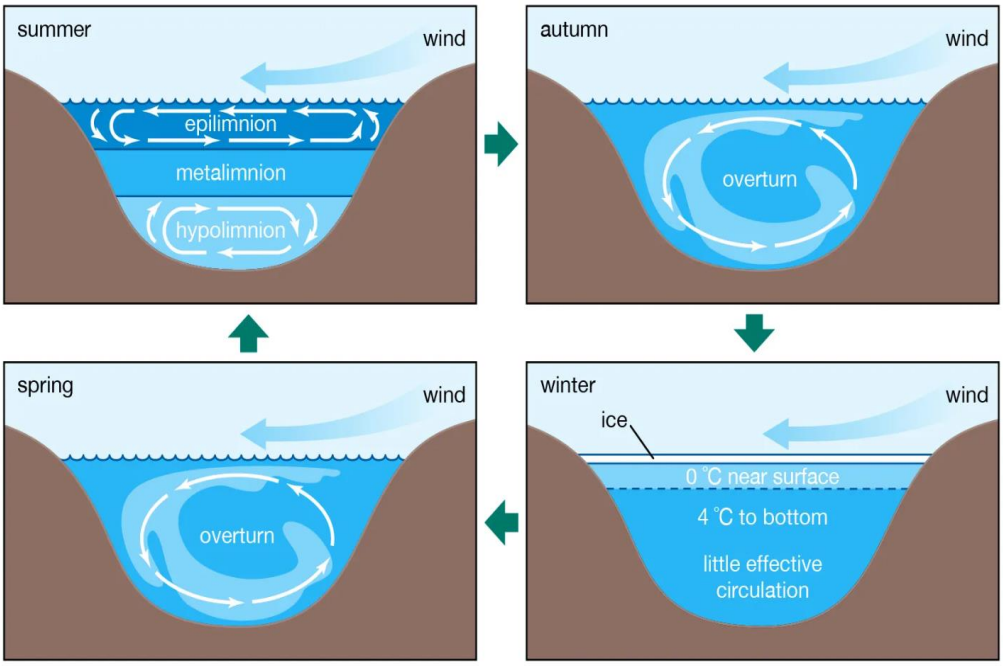

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Figure 1 The process of seasonal stratification occurring in deeper lakes (deeper than 4 meters)

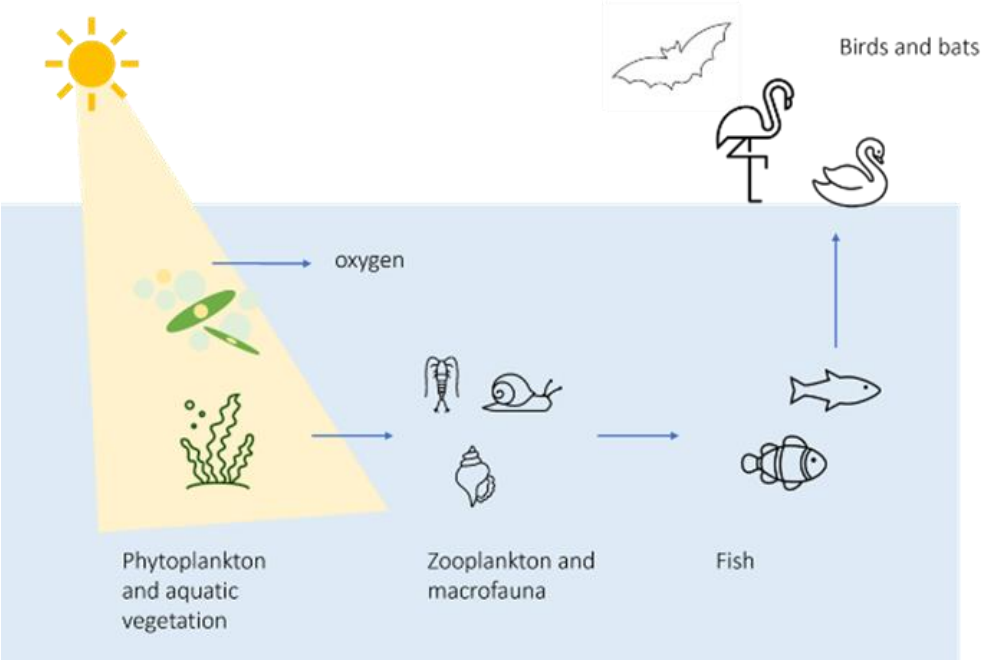

Figure 2 Simplified food web in a lake. The lower trophic levels include primary producers such as algae and aquatic vegetation, which utilize sunlight to produce oxygen and organic material. This organic material forms a food source for zooplankton and macrofauna. The remaining fauna (fish, birds and bats) shown in this picture is described as higher trophic level.

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On the short term, this document helps in choosing the right monitoring variables, as well as setting up a cost-efficient and effective monitoring plan, which is adequate for answering the questions asked by permitting authorities or other stakeholders. Furthermore, it helps creating consistency between datasets, which may then together form one large and internally consistent database. Creating such a database is important for a relatively new technology like FPV. With more knowledge, certain aspects of the relations between FPV design and impact on water types will transpire. This way the sector will learn how to balance FPV impact and performance. In the long term, this will reduce the need for intensive impact-monitoring. However, some minimum monitoring activities will remain relevant for every FPV plant despite the available knowledge, if only to categorize the waterbody or to identify new phenomena and trends over time.

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2 Finding FPV-suitable waterbodies

The implementation of FPV systems depends on finding suitable waterbodies. A waterbody is a certain clearly distinguishable part of surface water, such as a lake, a stream, river or a part of a river, lake or stream. The term waterbody also includes artificial reservoirs or smaller pools of water such as ponds. To identify FPV-suitable waterbodies, it is essential to consider their physical characteristics and their ecological sensitivity. Identifying these waterbodies relies on a combination of environmental assessments and data from robust sources.

2.1 Physical parameters

For the suitability of waterbodies to host FPV systems it is important to take into account the surface area, depth, hydrology and proximity to energy grids. Waterbodies with sufficiently large surface areas are generally more suitable for large-scale FPV systems, as they can better accommodate installations. This is due to economic feasibility, as larger installations often benefit from economies of scale and improved cost-effectiveness. However, a larger waterbody does not necessarily mean a lower ecological impact. The extent of the impact depends mainly on the percentage of the water surface covered by the panels. Information on the size of waterbodies within the Water Framework Directive (WFD) can be accessed through the Water Information System for Europe (WISE). Links to these databases are given below.

The depth of the waterbody can also play an important role. In deeper areas, there is typically not enough light at the bottom for growth of vegetation. As a result, installing FPV systems on these deeper parts reduces the risk of impacting aquatic vegetation (**Error! Reference source not found.**). Although specific size and depth requirements d epend on the project and local conditions, these characteristics play a key role in mitigating potential disruptions. Detailed depth data may require consultation of national databases. However, in some cases, depth information can also be derived from the WISE database, as waterbody types—such as those in the Netherlands—are linked to depth classifications.

Waterbodies with calm wave conditions often reduce stress on FPV infrastructure, simplifying deployment and maintenance. Data on wave dynamics can often be found in hydrological studies or basin-specific management plans. Proximity to energy grids and maintenance facilities is also crucial, as it directly affects economic feasibility and operational efficiency. Regional planning documents and infrastructure maps can offer insights into site accessibility.

2.2 Ecological sensitivity

Ecological sensitivity refers to the vulnerability of a waterbody and its ecosystems to disturbances, such as reduced light penetration and habitat alterations caused by floating photovoltaic (FPV) systems. Sensitive ecosystems, often characterized by the presence of unique or threatened species, critical habitats, or compromised conditions, are more susceptible to such disturbances. For example, a shallow pond with dense aquatic vegetation serving as a vital fish habitat. The introduction of FPV in such a

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pond, could disrupt both plant and animal life. Waterbodies with high ecological resilience—characterized by their capacity to withstand and recover from disturbances—exhibit lower ecological sensitivity and are more likely to endure FPV impacts with minimal disruption. In contrast, ecosystems with low resilience are more vulnerable to degradation and may experience irreversible changes. Key factors influencing resilience include water quality, habitat connectivity, and biodiversity. Thus, understanding ecological sensitivity is essential for evaluating the potential impacts of FPV systems and ensuring their placement minimizes ecological disruption.

Legal frameworks, such as the Water Framework Directive (WFD) and Natura 2000, provide tools to assess ecological sensitivity. The WFD classifies waterbodies based on their ecological and chemical status, using respectively Biological Quality Elements (BQE), which include fish, aquatic vegetation, macroinvertebrates, and phytoplankton and concentrations of chemical substances. Waterbodies classified as having “high”, "good" or "moderate" ecological status may be more resilient to changes, potentially making them suitable for FPV systems (Figure 3). Water authorities have the obligation to improve WFD waterbodies with bad or poor classification, these efforts may not be consistent with FPV installation. Under all circumstances authorities should prevent deterioration to a lower WFD class. Waterbodies within or close to Natura 2000 sites, protected under the EU Habitats and Birds Directives, have high ecological value and are subject to stringent regulations to prevent adverse impacts.

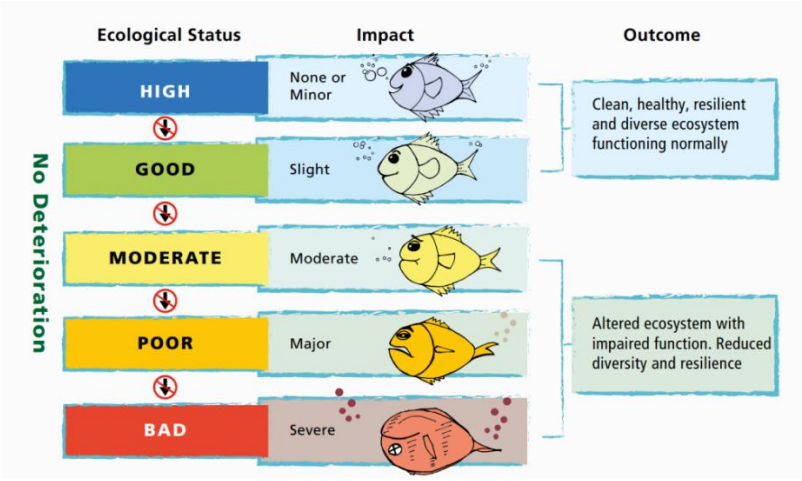

Figure 3 Illustration of the classification used in the Water Framework Directive which applies to all European waters. Picture from the Irish WFD website (www.catchments.ie).

By using these legal frameworks, we can effectively assess ecological sensitivity and ensure FPV systems are deployed in locations where they pose minimal ecological risks while supporting sustainable energy goals. There are different resources that can be used to evaluate the ecological conditions, legal status, and suitability of waterbodies for FPV deployment:

1. **WISE Water Framework Directive Database - European Environment Agency**
   The WISE WFD database contains data from the 1st and 2nd River Basin Management Plans reported by EU Members States according to article 13 of the Water Framework Directive (WFD). The database includes information

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about surface water bodies (number and size, water body category, ecological status or potential, chemical status, significant pressures and impacts). The information is presented by country, river basin district (RBD) and river basin district sub-unit (where applicable).

a. WISE Water Framework Directive (data viewer) | European Environment Agency's home page ; for an overview per country
b. Water Framework Directive - River Basin Management Plans | European Environment Agency's home page ; for a map view of the ecological or chemical status (figure 3)
c. DISCODATA ; the actual database with detailed information per WFD waterbody. To retrieve data, navigate to the WISE_WFD database, select the latest version, and open the SWB_SurfaceWaterBody table. Then, click on the three small dots next to the table and choose Open table viewer to view the data.

2. **Natura 2000 Viewer**
   The Natura 2000 Viewer provides access to data on protected areas under the Natura 2000 network, highlighting ecologically sensitive regions and habitats.
   a. Natura 2000 Viewer ; for a map view of Natura 2000 sites, habitats, species and countries.

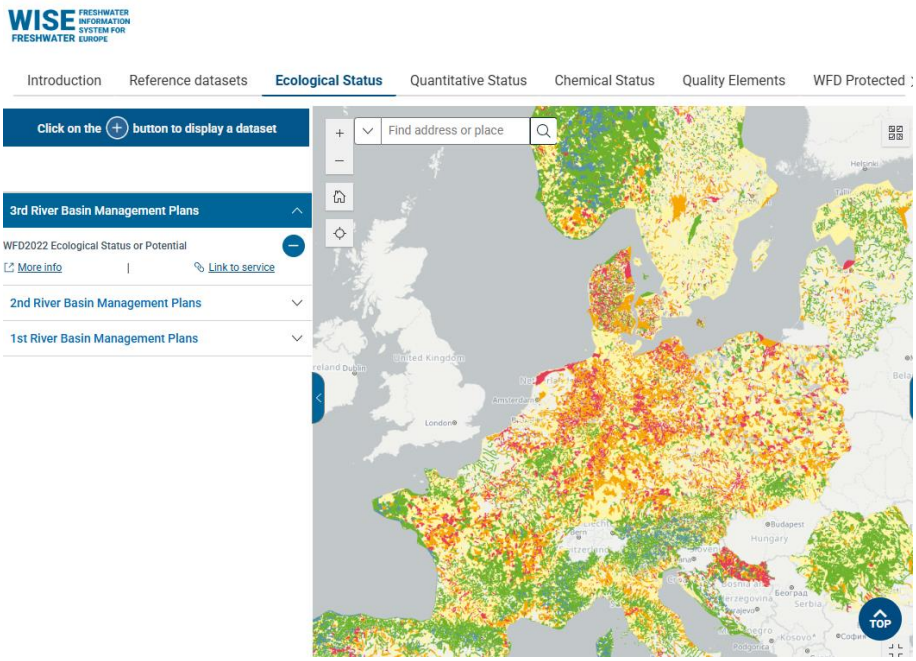

Figure 4 Map view of the ecological status of the 3rd (thus latest) River Basin Management Plans.

2.3 Non-WFD waterbodies

Some waterbodies fall outside the scope of the WFD and present viable opportunities for FPV installations. Artificial reservoirs and irrigation or industrial basins are not always classified as WFD waterbodies and often offer controlled environments with lower ecological sensitivity, making them favourable for development. Lists of artificial waterbodies can often be found in local water management plans or industrial zoning records. Similarly, brackish and intertidal zones, such as deltas and estuaries, may also offer opportunities for FPV systems, though these environments require additional

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consideration for wave conditions, salinity and tidal fluctuations. Coastal zone management plans and marine spatial planning frameworks provide relevant data for evaluating these areas.

In conclusion, identifying suitable waterbodies for FPV requires a comprehensive understanding of physical characteristics, ecological sensitivity, and the regulatory framework in place. The understanding can be supported by reliable data and collaboration with stakeholders. Resources such as the WISE system, Natura 2000 databases, and national planning inventories provide the tools necessary to align FPV site selection with ecological and legal standards, advancing renewable energy goals sustainably.

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3 Step by step towards monitoring plan

This chapter shows you the five steps for creating an effective monitoring plan. In short the five steps are:

1. **Understanding the waterbody of interest and intended FPV system.** Begin by analysing the characteristics of the waterbody to gain insight into its dynamics and baseline conditions.
2. **Identifying potential impacts and prioritizing parameters for the specific waterbody.** Determine the possible impacts of floating solar parks on the waterbody. Identify the most relevant indicators and variables that quantify these impacts.
3. **Defining where to place the monitoring locations.** Select the appropriate monitoring locations and determine the frequency and timing of measurements to ensure comprehensive data collection.
4. **Selecting Equipment and methods.** Choose suitable monitoring equipment based on the identified quantities to be measured and establish a realistic budget to support the monitoring activities.
5. **Database Management and Data Analysis.** Organize the collected data in a structured database and perform thorough data analyses to identify patterns and assess potential impacts.

This process is cyclical: findings from Step 5 may reveal the need for additional measurements or adjustments in the monitoring plan, ensuring an iterative approach to refining and improving the monitoring strategy.

3.1 Step 1: Understanding the waterbody of interest and intended FPV system

The first step in developing a monitoring plan for a FPV installation is conducting a comprehensive analysis of the considered waterbody. This involves gathering all available information on the characteristics of the waterbody where the FPV system is to be installed. Understanding the specific features and dynamics is critical, as the environmental impacts of the installation will depend significantly on the type of waterbody involved.

For instance, the expected impacts on an irrigation pond may differ substantially from those on a deep drinking water reservoir, a sandpit lake, or a large natural lake. Each waterbody type presents unique ecological, physical, and chemical conditions, which influence the potential effects of FPV installation and, consequently, the required

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monitoring strategies. The objective of this step is to collect existing data and knowledge about the waterbody to create a robust foundation for informed decision-making. By understanding the baseline conditions, a better prediction the potential environmental impacts of FPV installation can be made. The format provided in Appendix A allows you to easily fill in the details about the waterbody and the intended FPV system.

In addition to understanding the waterbody, it's also crucial to consider the properties of the potential FPV system you're thinking of, as the impact can vary significantly depending on factors like coverage percentage or the row distance between the panels (Appendix A).

3.2 Step 2: Identifying potential impacts and prioritizing parameters for the specific waterbody

Table 1 shows the recommended quantities for monitoring, henceforth called “Tier 1”. Figure 5 shows the indicators and the interactions in a simplified way. All Tier 1 quantities are those which are directly impacted by installing FPV (Figure 5). More information on these indicators will offer hydrologists and ecologist a solid base to assess the consequences for the waterbody and connected waterbodies.

*Table 1 showing the Tier 1 variables, to be measured at all cases. * Wind and hydrodynamics only in deeper lakes where stratification can occur (c. more than 5 meters deep)*

Indicator	Measuring what?	Frequency
Wind *	Windspeed and direction	Whole year round using in situ sensors
Hydrodynamics *	Waves	Whole year round using in situ sensors
Light in water column	Amount of available light	Whole year round using in situ sensors or monthly visits
Oxygen in water column	Concentration of oxygen	Whole year round using in situ sensors or monthly visits
Primary production	Chlorophyl-a	Whole year round using in situ sensors or monthly visits
Temperature in water column	Water temperature	Whole year round using in situ sensors or monthly visits
Turbidity [1]	Number of particles in water column	Whole year round using in situ sensors or monthly visits

[1] *This measures how clear or murky the water is. Important parameter to explain the available light in water column.*

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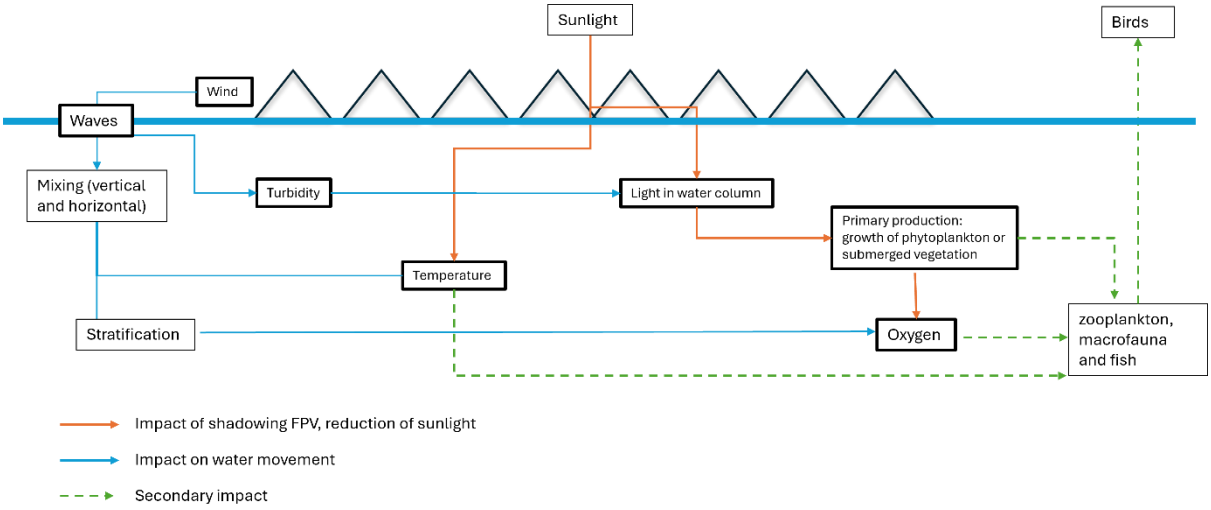

Figure 5 The interaction of the FPV and important indicators for flora and fauna in and nearby a waterbody.

In addition to Tier 1, Tier 2 monitoring quantities are quantities that depend on specifics of the location or can be required by permitting authorities or stakeholders. Some Tier 2 monitoring quantities are shown in Table 2. The table shows also the potential situations where these Tier 2 measurements might be necessary. Together with the stakeholders it can be decided to monitor variables for the Tier 2 list.

Please note that micro- or nanoplastics are not in the list of Tier 2 (Table 2). The reason is that we consider monitoring at the FPV plant (in situ) not an adequate method to determine if degradation of FPV components occurs over time. This should be examined in a laboratory setting.

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Table 2 showing the Tier 2 variables to be measured in some cases (please note that the order is not a priority list).

Indicator	Measuring what?	Frequency	When to be measured
Chemical composition (e.g., lead, antimony, PFAS[2])	Presence of relevant hazardous substances	Once a year. Or when a calamity like a fire has occurred	This could be required in a drinking water reservoir because very strict targets for concentrations of chemical substances apply. Could also be required when stakeholders are worried about certain substances like PFAS.
Nutrients and pH	Total N and P	6 times from April to October	Important to explain changes in productivity of the lake. Required in WFD waterbodies.
Submerged vegetation	Coverage functional groups and species	Once a year	Required when shallow parts[3] where submerged vegetation has potential to grow. Required in WFD waterbodies.
Phytoplankton	In addition to Chlorophyl-a (Tier 1), the composition of a population can be determined	Bi- weekly sampling from May to October	Can be required in waters sensitive to blue algae (eutrophic or mesotrophic). Measuring to substantiate the idea that algae composition changes, at the expense of blue algae. Required in WFD waterbodies.
Zooplankton	Density and species composition	Bi-weekly	Only required if there are worries about the impact on the overall food web. This will be indicated by the involved ecologist.
Fish	Species composition Biomass	Once a year	Substantiate the question if FPV has effect on fish stocks (+or -). Required in WFD waterbodies.
Organic buildup (flora and fauna growing on the FPV installation)	Species composition (algae, macrofauna, flora) Biomass estimates (weight)	Once a year	Can be required to substantiate if the buildup leads to increase of biodiversity or introduces invasive species. Can also be required for maintenance of anchor lines.
Birds	Number of breeders vs non-breeders. Number of indicators species	Monthly during breeding season For several years	Might be necessary in areas with rare bird species occurrences or nature protected areas. Required when the waterbody is in or close to Natura 2000 area (depending on the goals in that area).
Bats	Flight patterns	Frequency to be determined by involved specialists	Might be required in regions where bats are present and depend on this lake for food or when the lake's position is interfering with flyways. Required when the waterbody is in or close to Natura 2000 area (depending on the goals in that area).
Amphibians and reptiles	Numbers and species	Once every three years	Might be required in regions where rare species occur which might be impacted

[2] *Hazardous substances in solar panels can be lead and/or antimony (soldering materials). The backsheets of panels can contain PFAS. However, PFAS free solar panels are available as well.*

[3] *In most cases potential habitat for submerged vegetation is between 0 and 4 meters. However this range can be vary depending on lake characteristics.*

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Tier 3 monitoring quantities could be chosen to pursue research questions on the fringes of what is known about the FPV impact on waterbodies. For example, the impact of FPV covering on the exchange of (greenhouse) gases between water and atmosphere or the wearing of the used plastics in floaters. In most cases the questions will be initiated by research partners.

3.3 Step 3: Defining where & when

Floating solar parks can cover large parts of a waterbody. Where should one monitor and how many monitoring locations are needed? We present three options based on what we learned from scientific papers and on general knowledge of measurement strategies used in aquatic ecological field research (mainly the Water Framework Directive). Feasibility is also an aspect which is considered for the options. After the options for location we show you some options for frequency and duration.

Option 1 –reference point in a similar waterbody
This option requires a second identical or at least comparable waterbody to act as a reference (Figure 6). Waterbodies are comparable if they have about the same size, depth, bottom sediments and are fed by the same source (e.g., river or groundwater). This is the preferred option. To find a situation like this will be however, rare.

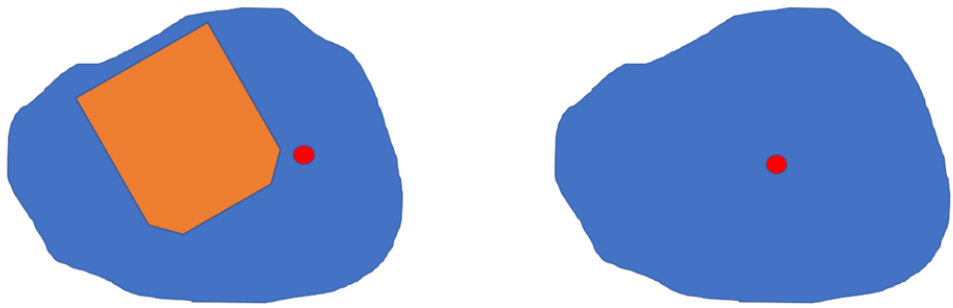

Figure 6 Monitoring of a waterbody with (left) and without FPV system (right). The red dot is the measurement location.

Option 2 – reference point in same waterbody as FPV
In case there is no other waterbody which can act as a reference, the advice is to choose two monitoring locations in the same waterbody: one under and one outside the PV system (Figure 7). The location outside the FPV installation acts as the reference site. This reference site should be chosen such as to be minimally affected by the presence of the FPV installation, while still being representative of the waterbody portion covered by the FPV.

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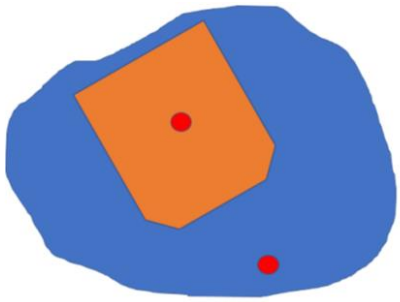

Figure 7 Red dots are monitoring locations. The point outside the FPV system must be chosen in such a way that it is outside the sphere of influence of the FPV system but not too close to the shores

Option 3 – choose a gradient

This option involves monitoring at different locations on and beyond the FPV installation (Figure 8) to spatially resolve the FPV impact on the waterbody. Monitoring results will show whether there are differences between the centre of the PV system and the edge and how the effects change as the distance from the PV system increases. In the sketch of Figure 8, three measuring points are drawn, but there could of course be more.

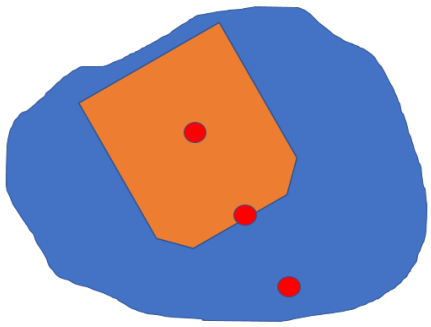

Figure 8 Choosing a gradient in monitoring in a waterbody with a PV system. Red dots are monitoring locations.

After the location is chosen, the monitoring frequency and duration need to be chosen. Tables 1 contains the frequency of monitoring for the Tier 1 variables, duration is at least one year to include all seasonal changes. Continuous monitoring can be done by placing sensors in the water at a fixed depth (at c. 80 to 100 cm) or along a vertical string to capture also the changes against depth. These sensors will provide information of the surrounding water conditions every hour. If installations of equipment for continuous measurements is not possible an alternative is visit the FPV each month and perform the measurements during the visit (Figure 9).

In short, we have three options regarding frequency. All three option will provide you with sufficient data to assess the impact. The choice depends on the available budget, feasibility, and wishes from associated researchers, permitting bodies or other stakeholders. The three are:

1. continuous measurements against depth for all Tier 1 variables
2. continuous measurements at fixed depths for all Tier 1 variables
3. monthly visits for all Tier 1 variables

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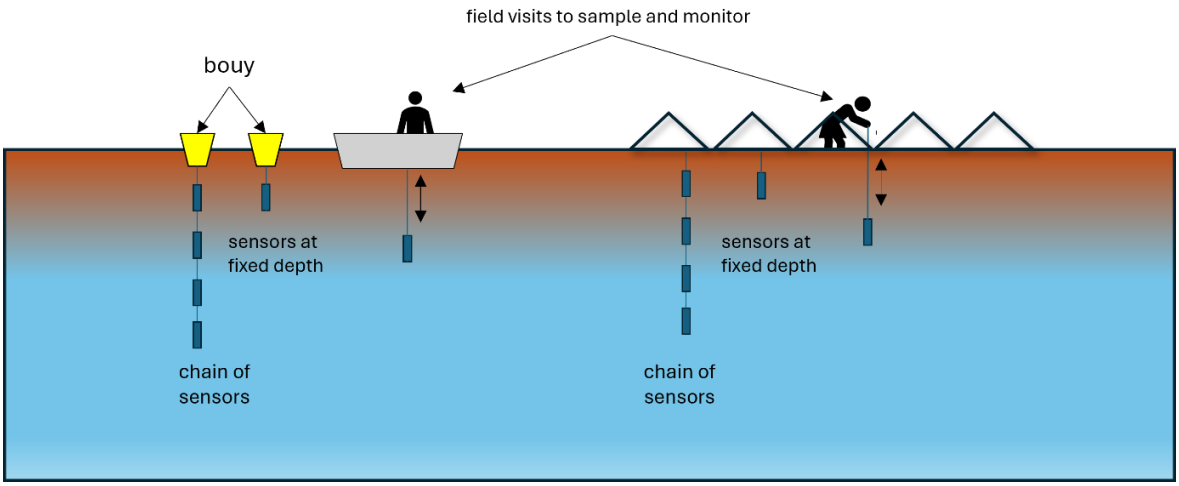

Figure 9 Schematic illustration of the different options regarding the use of sensor at fixed stations or monthly visits.

3.4 Step 4: Selecting equipment and methods

This step helps to choose suitable monitoring equipment based on the identified quantities to be measured and establish a realistic budget to support the monitoring activities (Table 3). The information on prices is indicative and the mentioned equipment is not exhaustive. We advise to request quotes at local specialized monitoring companies.

Table 3 showing the possible equipment that can be used for Tier 1 and 2 measurements.

Indicators	Measured variable	Measuring Device	Purchase costs (order of magnitude)
Tier 1			
Hydrodynamics	Water movement	Wave Buoy	High, price per buoy ~ 9.000 EUR
Wind	Windspeed and direction	Anemometer	Low, price per sensor ~ 900 EUR
Light in water column	Amount of available light	PAR sensor for continuous measurements	Low, price per sensor ~ 500 EUR
		Secchi disk for one time measurements	Very low to purchase (~20EUR) but needs personnel cost
Oxygen in water column	Concentration of oxygen	Optical (Luminescent) DO Sensor Chain (continuous at various depths)	Medium, price per chain ~ 5.000 EUR
		Exo YSI optical dissolved smart sensor. (continuous at fixed depth)	Low, price per sensor ~ 2500 EUR
Primary production	Chlorophyl-a	Optical Fluorescence Sensor (Exo YSI Sensor)	Low, price per sensor ~ 1.500 EUR
Temperature in water column	Water temperature	Thermistor Chain (continuous at various depths)	High, price per chain ~ 10.000 EUR
		Exo YSI sensor (continuous at fixed depth)	Low, price per sensor ~ 2500 EUR

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Turbidity	Number of particles in water column	Exo YSI sensor (continuous at fixed depth)	Low, price per sensor ~ 2500 EUR
Multiprobe including most Tier 1 variables	light, temperature, pH, redox potential, dissolved oxygen, oxygen saturation, salinity, turbidity, conductivity, and chlorophyll-a	A fixed-position multiparameter probe Aquaread 'AP7000	High, price ~ 15.000 EUR (however, all variables are included)
Tier 2			
Chemical composition	Concentrations or bioassays[4]	Sampling and analysing in laboratory	Depend on number of samples. E.g., analysis of PFAS is estimated to be c. 700 EUR per sample in a certified laboratory. Additional costs of field sampling
Nutrients	Total	Sampling and laboratory analysis	Low costs per sample
Submerged vegetation	Coverage of groups and species	Observations	Mostly only along the shores in shallow areas. Observations from the shore or a boat are sufficient. No diving is required. Follow the national protocols for monitoring (WFD). Hiring a specialized consultant is a good option.
Phytoplankton	Species and abundance	1Sampling and determination 2 FRRF (Fast Repetition Rate Fluorometry)	1High due to laboratory work and the high number of species 2. used for estimation of phytoplankton abundance
Zooplankton	Species and abundance	Sample probing in laboratory	High due to laboratory work. Costs are up to 200 EUR per sample, mostly lower than phytoplankton determination.
Fish	species	Underwater Cameras/field sampling/electrofishing	Low-High depending on method. Expert costs for field sampling and electrofishing. Hiring an expert consultant is a good option.
Birds	(indicator) species	Cameras or field observations	Low-Medium depending on method and intensity. Expert costs for field observations and analyses of the camera results. Hiring a expert consultant is a good option.
Bats	species	Bat detectors and field observations	Low-Medium depending on method and intensity. Expert costs for field sampling and analyses of noises form detectors. Hiring a expert consultant is a good option.
Organic build up	Species and biomass	Field sampling, weighing, and determination	

Tier 1 measurements are taken continuously with sensors while tier 2 measurements are taken with probing or sampling. The preparation and installation of continuous sensors typically takes 3 to 4 days with qualified personnel. Maintenance requirements

[4] bioassays are tests that use living organisms, cells, or tissues to measure the effects of a substance

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are usually low; however, unexpected issues can lead to significant cost increases if additional trips to the location become necessary.

For sampling of non-continuous measurements, a full working day is required per parameter. Travel expenses back and forth as well as the daily rate of expert personnel rate need to be taken into account. The probing/sampling frequency of non-continuous parameters depends on the scope of the measurement and the conditions at the location. For example, bird analysis should be scheduled in accordance with bird migration patterns, if applicable to the location in question. As a rule of thumb, 4-5 observation sessions yearly are required.

The data evaluation in the case of non-laboratory analysis can typically be completed within one working day, provided experts and automated systems are utilized. Analysis involving laboratory processing, as it can be necessary for phytoplankton or zooplankton, is more costly. Diving investigations, due to the need for equipment and skilled personnel, result in higher costs as well, whereas bird monitoring is generally more affordable. The established analysis of fish populations using electrofishing is considered expensive due to specialized equipment and expertise requirements.

3.5 Step 5: Database management and data analysis

One of the goals of developing a monitoring strategy for assessing the potential effects of floating solar systems on water quality and ecology is to gain a general understanding of these effects, rather than focusing on a single specific system. This broader knowledge can also support organizations in showcasing their expertise internationally.

Currently, there is no standardized format or central repository for storing data on FPV systems and their environmental effects. As a result, valuable information is often fragmented, making comparisons and large-scale analyses challenging. To address this, it is essential that all measurements are stored in a structured, standardized, and centrally accessible manner for professionals working on this topic. This approach ensures secure and centralized data storage while keeping information easily accessible for research, policy-making, and system optimization.

Our advice for storing the collected data in a structured and accessible manner is a relational database (e.g., PostgreSQL, SQLite) or a well-organized CSV/Parquet file structure if working with simpler data management tools. Then the following aspects of the obtained data are recommended (also Table 5):

1. Store data in long format:
    a. Each row represents a single observation (i.e., one parameter at one time point).
    b. This makes data handling and analysis more simple.
2. Standardized naming:
    a. Use clear, consistent parameter names (e.g., water_temp instead of WT).
    b. Use lowercase and underscores for readability (e.g., wind_speed).

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3. Metadata Focus:
    a. Include metadata columns to describe the data
    b. Important metadata fields:
        i. location_id (Unique site identifier)
        ii. measurement_type (below FPV sytem, next to FPV system, in open water)
        iii. fpv_system_type (Floating PV system classification)
        iv. coverage_percent (Percentage of water surface covered by FPV)
        v. date_time (Timestamp in ISO 8601 format)
        vi. parameter_name (Measured variable)
        vii. value (Observed measurement)
        viii. unit (Unit of measurement)
        ix. data_source (Sensor/manual sampling)

Table 4 Example data frame of monitoring effort in a long format.

location_id	measurement_type	fpv_system_type	coverage_percent	date_time	parameter_name	value	unit	data_source
A01	below_FPV	Type_A	30	2025-02-14 10:00:00	water_temperature	15.2	°C	Sensor_X
A01	next_to_FPV	Type_A	30	2025-02-14 10:00:00	dissolved_oxygen	8.5	mg/L	Sensor_Y
A02	open_water	Type_B	45	2025-02-14 11:00:00	chlorophyll	3.2	µg/L	Lab Analysis
A02	below_FPV	Type_B	45	2025-02-14 11:00:00	pH	7.8		Sensor_Z
A03	next_to_FPV	Type_C	60	2025-02-14 12:00:00	turbidity	2.5	NTU	Manual Sampling

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4 Baseline measurements

Regardless of which monitoring strategy is chosen, it is recommended to always carry out baseline measurements; meaning a campaign before installing the PV systems. This is best done the year before. In some cases baseline information is available. However, in most cases it won't.

Once a waterbody has been sufficiently assessed and deemed suitable for the installation of an FPV system (step 1), it is essential to establish a baseline understanding of its ecological and physical conditions. Baseline measurements are critical for evaluating the impact of the FPV system post-installation. These measurements should focus on parameters that directly correspond to the primary environmental pressures introduced by FPV systems. Our advice is to monitor all quantities that reflect the most direct impacts of FPV on the waterbody through shading and wind shear, the so called Tier 1 variables (Table 1).

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5 Innovations in monitoring techniques

5.1 (Underwater) Drones

Underwater drones are being increasingly used in water management. Very often the purpose is simply inspection, but the drones are nowadays also equipped with sensors to monitor water quality variables such as temperature and oxygen. Monitoring under floating PV systems has also already been undertaken (Pedroso et al., 2021). An important advantage of drones is that they can travel to locations (both horizontally and vertically) under a solar park that otherwise would be difficult to reach by conventional monitoring. This way, 3D-resolved fluctuations of certain water quality parameters can be obtained. A big challenge is to keep the drone in position to be able to perform the measurements at selected locations.

5.2 Remote sensing

Earth observations can be used to monitor several water quality parameters such as chlorophyll-a, suspended matter and yellow matter. A major advantage is their coverage of large areas allowing therefore to observe complete waterbodies. Satellites suitable for monitoring water quality have a revisit period of several days (Sentinel-2 for example has a revisit time of 5 days) which means that every 5 days a new satellite image is available to be processed, assuming that there is no cloud coverage. There are limitations, however. One of them is the size of the waterbody which needs to be considered in relation to the spatial resolution of the sensing instrument on board the satellite. Sentinel-2 has a spatial resolution of 10, 20 and 60 m. Small waterbodies are therefore problematic to be monitored with satellites. Cloud coverage is another limitation because clouds block the satellite's sensors from directly observing the water surface beneath them, they may scatter sunlight in various directions, creating atmospheric interference and cloud shadows can distort the light being reflected off the water. Finally, not all water quality parameters can currently be (directly) monitored with earth observations. Despite the limitations, earth observation data can be used to monitor the impact of floating solar parks on water quality in case waterbodies are large enough.

Currently, new Sentinel expansion missions are being developed (ESA). Some of these missions will come with more spectral bands (hyperspectral) and improved spatial resolutions. For water quality, especially the CHIME mission will be of interest (CHIME).

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6 Appendix A – Information sheet for step 1

Characteristic description	Properties
Name of the waterbody	
Location of the waterbody (x, y coordinates)	
Size/area of the waterbody	
Maximum depth of the waterbody	
Bathymetric chart	
Hydrological connection with surrounding water (e.g. isolated, connected to river, connected to canal)	
Predominant wind direction	
Ecosystem services of the waterbody (e.g. recreation, transport, nature)	Water usage east: Water usage South: Water usage west: Water usage north:
Is waterbody classified as a WFD waterbody or in a protected nature reserve (e.g. Natura 2000 area)?	
Current ecological and chemical status (by WFD classification or expert judgment)	
What measurements are available? (e.g. nutrients, oxygen, temperature, bird observations, fish counting's, bat observations or other biological data	
Solar park name	
Initiator/founder	
Contact details	
(Expected) year of installation	
Number of solar panels	
Total area of the solar park (in hectares)	
Area covered by solar panels (in percentage)	
Border width to solar panels	
Row distance between solar panels	
Number of rows with solar panels	
Row length	
Orientation of solar panels	
Panels fixed or tracking the sun?	
Type of anchoring (land, shore, seabed)	
Minimum height (in meters)	

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Maximum height (in meters)	
Horizontal gaps	
Vertical gaps	
Space between floats and solar panels	
Glass plates present?	
Type of material for floating components	

7 Appendix B- Example monitoring plan

Following the steps of this guideline, a monitoring plan for a typical waterbody and respective FPV system is shown as an example in the following. The primary goal of this monitoring plan is to evaluate the ecological and physical-chemical impacts of the FPV system. The required sensor quantity is mostly independent of the lake and FPV size, as the FPV influence is analysed with the relation between critical points of interest (e. g. plant centre and reference area).

Tier 1: Continuous monitoring

- **Light in the water column (PAR sensors)**

  Locations: Below FPV-System (1 x in the middle, 1 x at the edge, 1 x in-between) and 1 x in reference area outside FPV system influence

  Cost: Approximately 500 EUR per sensor, i. e. 2.000 EUR

- **Dissolved oxygen (DO sensor chain)**

  Locations: 1 x below FPV system, 1 x in reference area

  Cost: Approximately 5.000 EUR per chain, i. e. 10.000 EUR

- **Chlorophyll-a for phytoplankton growth (optical fluorescence sensor)**

  Locations: 1 x below FPV system, 1 x in reference area

  Cost: Approximately 1.500 EUR per sensor, i. e. 3.000 EUR

- **Water temperature (thermistor chain)**

  Locations: 1 x below FPV system, 1 x in reference area

  Cost: Approximately 10.000 EUR per chain, i. e. 20.000 EUR

Installation Requirements: 3–4 days for deployment with qualified personnel (estimated cost of 5,000 EUR)

Maintenance: Estimated at 2,000 EUR annually. No regular maintenance is required/scheduled to begin with, but it is expectable that software and data

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management as well as sensor troubleshooting, including the corresponding travels, leads to expenses.

8 References

Pedroso de Lima R.L., K. Paxinou, F. Boogaard, O. Akkerman, F. Lin, 2021. In-SituWater Quality Observations under a Large-Scale Floating Solar Farm Using Sensors and Underwater Drones. Sustainability 2021, 13, 6421. https://doi.org/10.3390/su13116421

9 List of Acronyms

- SuRE: Sustainable, Reliable and Efficient Floating Photovoltaic Power Plants
- WFD: Water Framework Directive
- BQE: Biological Quality Elements
- RBD: River Basin District
- ESA: European Space Agency
- CHIME: Copernicus Hyperspectral Imaging Mission for the Environment
- FPV: Floating Photovoltaic

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